📚 OnamVPN Complete Project Documentation

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Project Overview

OnamVPN is a modern, cross-platform VPN application built with Python and PySide6. It provides a graphical user interface for managing VireGuard VPN connections with support for multiple servers, real-time ping testing, and comprehensive logging.

Key Features

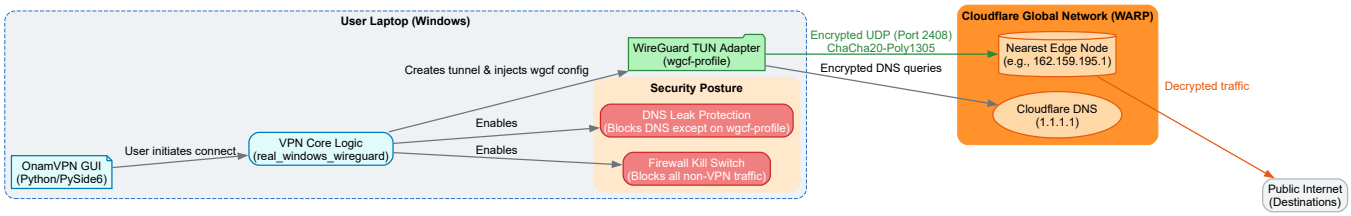
- ☑ **Cross-Platform GUI**: Works on Windows, Linux, and macOS with PySide6/Qt
- ☑ **WireGuard Integration**: Leverages WireGuard for secure VPN tunneling
- ☑ **Multiple Handlers**: Platform-specific implementations for optimal performance
- ☑ **Real-time Monitoring**: Continuous connection status and ping testing
- ☑ **Fallback Mode**: Works even without WireGuard installed (demo mode)
- ☑ **Server Management**: Support for multiple VPN servers with geolocation
- ☑ **Comprehensive Logging**: Rotating file logs with configurable verbosity
- ☑ **Settings Persistence**: User preferences saved in JSON format

Technology Stack

Component	Technology
GUI Framework	PySide6 (Qt for Python)
VPN Protocol	WireGuard
Encryption	Cryptography library (AES)
Logging	Python logging + RotatingFileHandler
Threading	Python threading + Qt QThread

Component	Technology
Configuration	JSON
Package Manager	pip

System Architecture



Architecture Layers

The application follows a **layered architecture** pattern:

1. GUI Layer (Presentation)

- **MainWindow** (`gui/main_window.py`) - Primary UI controller
  - Manages window lifecycle
  - Handles user interactions
  - Updates connection status
  - Manages theme and language switching
- **ServerGrid** (`gui/server_grid.py`) - Server selection interface
  - Displays available servers as interactive cards
  - Shows real-time ping latency
  - Handles server selection
  - Performs concurrent ping testing
- **SettingsPanel** (`gui/settings_panel.py`) - Configuration dialog
  - Theme/language selection
  - User preference persistence
  - Settings validation
  - About/help information

2. VPN Core Layer (Business Logic)

- **WireGuardHandler** - Cross-platform handler (Linux/macOS)
  - Direct WireGuard CLI integration
  - Key generation and management
  - Config file creation
  - Subprocess-based tunnel control

- **RealWindowsWireGuard** - Windows-specific handler
  - Windows path detection
  - WireGuard app integration
  - User-guided configuration flow
- **SimpleVPNHandler** - Demo/fallback handler
  - No WireGuard required
  - Simulated VPN behavior
  - Educational mode

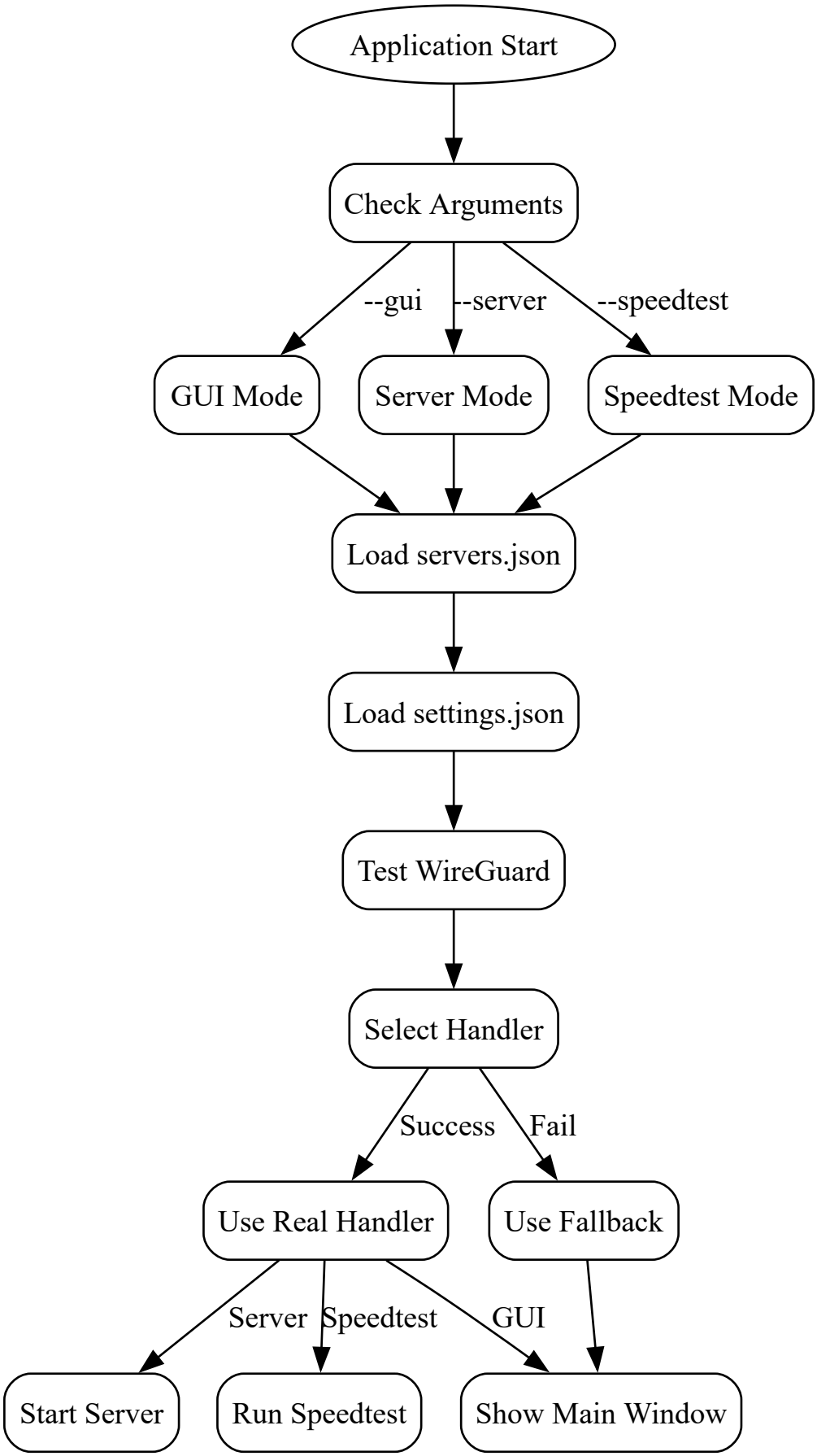
3. Utilities Layer

- **Logger** - Centralized logging system
  - File rotation (10MB per file, 5 backups)
  - Console + file output
  - Cross-platform log directory
- **SpeedTest** - Ping and speed testing
  - TCP socket connectivity test
  - ICMP ping measurement
  - Concurrent testing with ThreadPool
- **Encryption** - Cryptographic operations
  - AES encryption/decryption
  - Password hashing
  - Key derivation

4. Data Layer (Persistence)

- **servers.json** - Server configuration
  - List of available VPN servers
  - Server metadata (location, country, endpoint)
  - Public keys for WireGuard
- **settings.json** - User preferences
  - Theme (light/dark)
  - Language selection
  - Window geometry
  - Connection history
- **Config Files** - WireGuard configurations
  - Client `.conf` files
  - Server `.conf` files
  - Timestamped backups

Application Startup Flow



Startup Sequence

```
1. main.py::__main__()
   ├─ Parse command-line arguments
   ├─ Determine application mode (GUI/Server/Speedtest)
   │
   ├─ Load Configuration
   │  ├─ Initialize logger
   │  ├─ Load servers.json
   │  └─ Load settings.json
   │
   ├─ Detect Platform & Handler
   │  ├─ Check system: Windows/Linux/macOS
   │  ├─ Test WireGuard installation
   │  └─ Select appropriate handler
   │
   └─ Launch Mode
      ├─ GUI Mode: Show MainWindow
      ├─ Server Mode: Start VPN server
      └─ Speedtest Mode: Run speed tests
```

Handler Selection Logic

```python
if platform.system() == 'Windows':
    try:
        handler = RealWindowsWireGuard()
        if handler.test_wireguard_installation():
            vpn_handler = handler  # Real handler
        else:
            vpn_handler = SimpleVPNHandler()  # Fallback
    except:
        vpn_handler = SimpleVPNHandler()  # Fallback
else:  # Linux/macOS
    try:
        handler = WireGuardHandler()
        if handler.test_wireguard_installation():
            vpn_handler = handler  # Real handler
        else:
            vpn_handler = SimpleVPNHandler()  # Fallback
    except:
        vpn_handler = SimpleVPNHandler()  # Fallback
```

Mode Details

GUI Mode (Default)

- Displays MainWindow with server grid
- Monitors connection status every 2 seconds
- Handles user interactions
- Saves settings on exit

Server Mode (`--server`)

- Starts a VPN server instead of client
- Listens on port 51820 (configurable)
- Accepts client connections
- Manages tunnel encryption/decryption

Speedtest Mode (`--speedtest`)

- Runs speed tests against all servers
- Measures ping latency
- Reports results in CSV format
- Useful for benchmarking

VPN Connection Lifecycle

Cloudflare WARP Tunnel Creation

Phase 1-3: Initialization (< 1 second)

1. **User clicks "Connect"** on ServerCard
2. **Server selected** and validated
3. **WireGuard checked** for availability

Phase 4-6: Key Generation & Config (2-5 seconds)

4. **Private/public keypair generated**
   - Uses `wg genkey` command
   - Secure asymmetric cryptography
5. **WireGuard config created**
   - Interface section (address, DNS)
   - Peer section (server endpoint, public key)
   - Persistence settings
6. **Config file saved**
   - Platform-specific location
   - Read-only permissions (0600)

Phase 7-9: Thread & Tunnel Setup (10-30 seconds)

7. **Background thread started**
   - Offload blocking operation
   - Keep UI responsive
   - Non-blocking from user perspective

8. `wg-quick up` **executed**
    - Creates TUN/TAP interface
    - Configures routing rules
    - Establishes peer connection
9. **Tunnel established**
    - VPN interface active
    - Traffic routed through tunnel
    - Persistent keepalive enabled (25s)

Phase 10-12: Finalization (< 1 second)

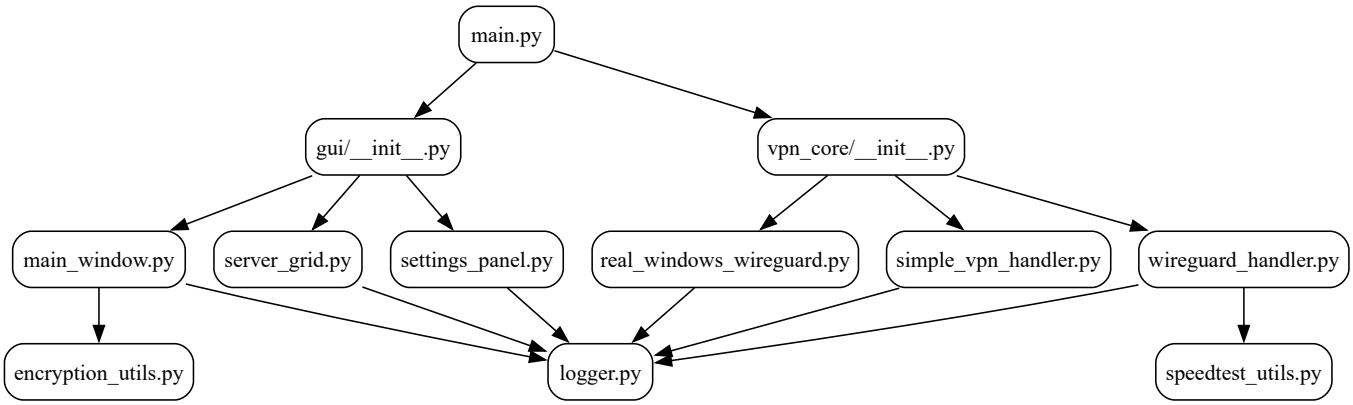
10. **Connection state updated**
    - `handler.is_connected = True`
    - `handler.current_server = server`
11. **UI refreshed**
    - Status label updated
    - Connect button disabled
    - Disconnect button enabled
    - Server grid refreshed
12. **User sees "Connected" status**
    - VPN fully operational
    - Ping tests continue
    - Logs recorded

On Disconnect

```
User clicks "Disconnect"
    ↓
handler.disconnect()
    ↓
Execute: wg-quick down [config_file]
    ↓
Delete temporary keys
    ↓
Update UI state
    ↓
Log completion
```

Disconnect Duration: 5-10 seconds

Module Dependencies



Import Hierarchy

```
main.py (Entry Point)
├─ gui/
│  ├─ main_window.py
│  │  └─ ServerGrid, SettingsPanel
│  ├─ server_grid.py
│  │  └─ SpeedTest utilities
│  └─ settings_panel.py
│     └─ JSON persistence
│
└─ vpn_core/
   ├─ wireguard_handler.py
   ├─ real_windows_wireguard.py
   ├─ simple_vpn_handler.py
   ├─ speedtest_utils.py
   ├─ logger.py
   └─ encryption_utils.py
```

External Dependencies

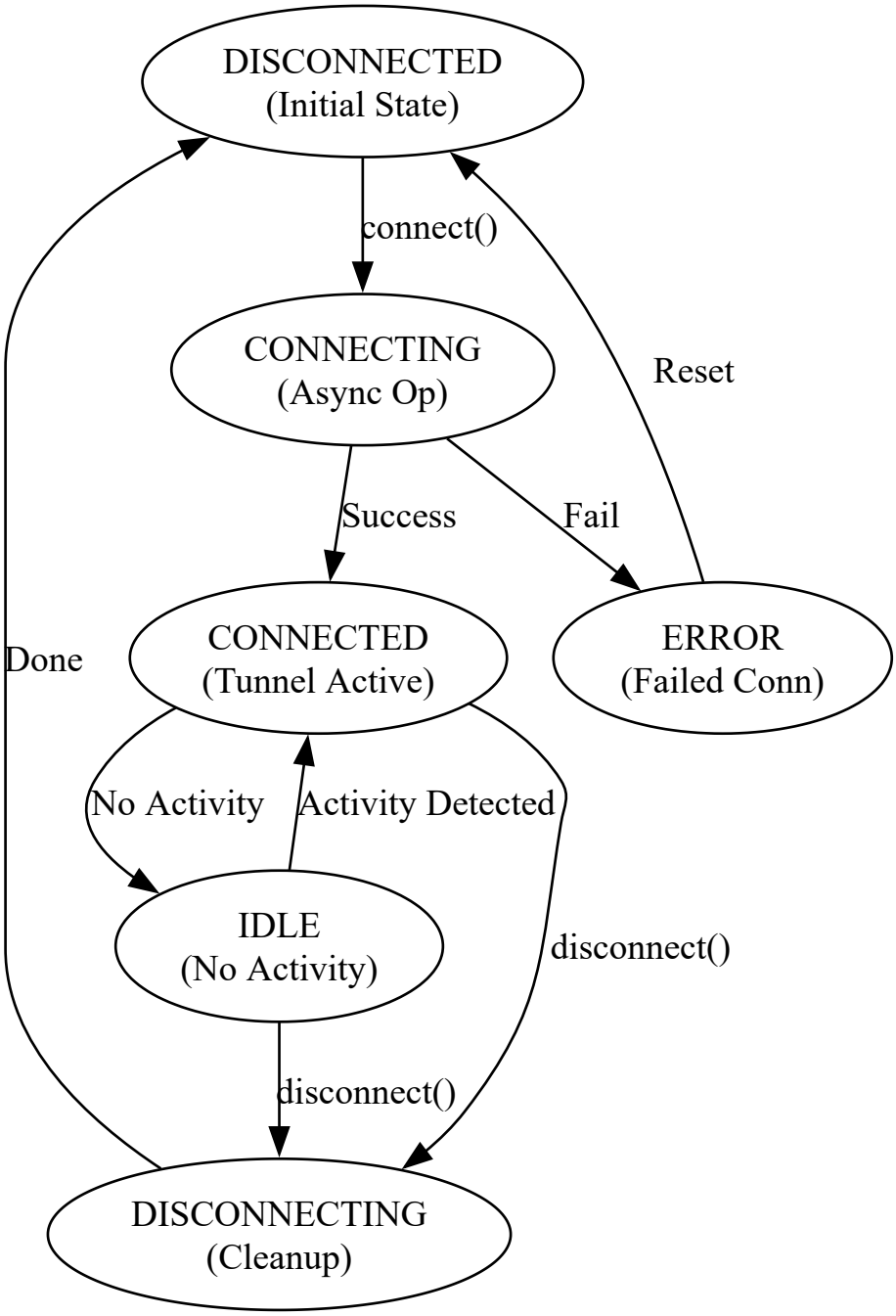
```
PySide6            # Qt GUI framework
cryptography       # AES encryption
paramiko           # SSH (optional)
requests           # HTTP client (optional)
```

Internal Dependencies

Module	Depends On	Purpose
main.py	gui, vpn_core	Entry point, mode selection
MainWindow	ServerGrid, SettingsPanel, handlers	UI orchestration
ServerGrid	SpeedTest	Ping display, selection
SettingsPanel	logger	Settings persistence

Module	Depends On	Purpose
WireGuardHandler	logger, speedtest	VPN operations
RealWindowsWireGuard	logger	Windows-specific VPN
SimpleVPNHandler	logger	Demo mode
SpeedTest	logger	Ping testing
Logger	logging	Centralized logging
Encryption	cryptography	Crypto operations

Connection State Machine



State Definitions

DISCONNECTED (Initial State)

- No active VPN tunnel
- Handler ready to accept connections
- Default state after startup

CONNECTING (Transition)

- User initiated connection
- Background operation in progress
- UI shows loading spinner
- User cannot interact with connection controls

CONNECTED (Stable State)

- VPN tunnel fully established
- `wg show` confirms tunnel active
- Connection status displayed
- Ping tests continue
- User can disconnect at any time

IDLE (Substates)

- Connected but no recent activity
- Keepalive packets still sent
- Monitoring continues
- Timeout triggers if > 5 minutes idle

DISCONNECTING (Transition)

- User requested disconnection
- Cleanup operations in progress
- UI shows status

ERROR (Terminal)

- Connection attempt failed
- Error reason logged
- User can retry or select different server
- Automatic fallback to DISCONNECTED after reset

State Transitions

From	To	Trigger	Duration
DISCONNECTED	CONNECTING	`connect()`	< 1s
CONNECTING	CONNECTED	Success	10-30s
CONNECTING	ERROR	Failure	1-30s

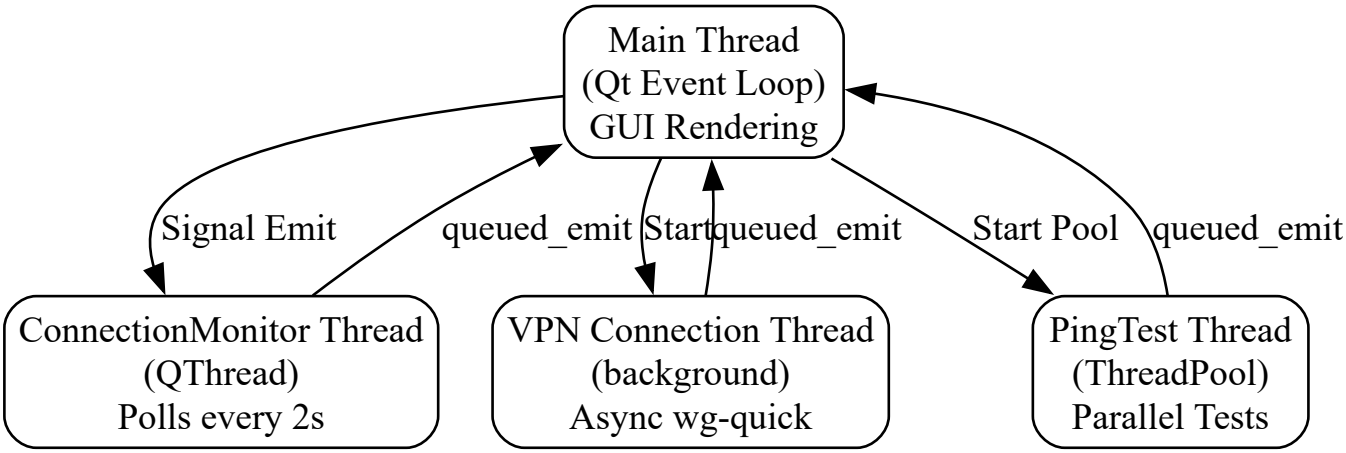
From	To	Trigger	Duration
CONNECTED	IDLE	Inactivity	5m+
IDLE	CONNECTED	Activity	< 1s
CONNECTED/IDLE	DISCONNECTING	`disconnect()`	< 1s
DISCONNECTING	DISCONNECTED	Complete	5-10s
ERROR	DISCONNECTED	Reset/Retry	< 1s

State Machine Code Location

File: `vpn_core/wireguard_handler.py`

```
class ConnectionState(Enum):
    DISCONNECTED = "disconnected"
    CONNECTING = "connecting"
    CONNECTED = "connected"
    IDLE = "idle"
    DISCONNECTING = "disconnecting"
    ERROR = "error"
```

Threading Model



Thread Architecture

```
┌──────────────────────────────────────────────────────────┐
│                    Qt Event Loop                         │
│             (Main Thread - GUI Rendering)                │
│                                                          │
│  • UI event handling                                     │
│  • Signal/slot processing                                │
│  • OpenGL rendering                                      │
│  • NOT for blocking operations                           │
```

```
└──────────────┬──────────────────────────────────────────┘
               │
               ├──────────────────────────────────────────┐
               │                                          │
        ┌──────▼──────────┐                ┌──────────────▼──────────┐
        │ ConnectionMonitor│                │ VPN Connection          │
        │   (QThread)      │                │ Thread (background)     │
        │                  │                │                         │
        │ • Polls every 2s│                │ • Async wg-quick        │
        │ • Checks status  │                │ • Key generation        │
        │ • Emits signals  │                │ • Config creation       │
        │ • Queued emit    │                │ • Subprocess calls      │
        │   to main        │                │ • Queued signal to      │
        │                  │                │   main thread           │
        └──────────────────┘                └─────────────────────────┘
               │
               │
        ┌──────▼──────────────────────────────────────────┐
        │    PingTest Thread Pool                         │
        │    (concurrent.futures)                         │
        │                                                 │
        │ • Worker 1: Ping Server 1                       │
        │ • Worker 2: Ping Server 2                       │
        │ • Worker 3: Ping Server 3                       │
        │ • Worker 4: Ping Server 4                       │
        │ (up to 8 parallel pings)                        │
        │                                                 │
        │ Results aggregated, queued to main              │
        └─────────────────────────────────────────────────┘
```

Thread Specifications

Main Thread (Qt Event Loop)

- **Responsibility:** GUI rendering, user interaction
- **Duration:** Entire application lifetime
- **Blocking Operations:** None (all async)
- **Communication:** Receives signals from other threads

Code Location: `gui/main_window.py`

ConnectionMonitor Thread

- **Type:** QThread (Qt-aware threading)
- **Interval:** 2-second polling cycle
- **Operation:** Calls `wg show` to check tunnel status
- **Signal:** Emits `connection_status_changed` signal (queued)
- **Thread Safety:** Qt's signal/slot mechanism (thread-safe)

Code Location: `gui/main_window.py::ConnectionMonitor`

```
class ConnectionMonitor(QThread):
    status_changed = pyqtSignal(dict)  # Queued signal

    def run(self):
        while self.is_running:
            status = self.handler.get_connection_status()
            self.status_changed.emit(status)  # Queued delivery
            time.sleep(2)
```

VPN Connection Thread

- **Type:** `threading.Thread` (background worker)
- **Trigger:** When user clicks "Connect"
- **Operation:** Blocks on `wg-quick up` (10-30 seconds)
- **Signal:** Sends custom signal when done
- **Cleanup:** Daemon thread, killed on app exit

Code Location: `vpn_core/wireguard_handler.py::_connect_thread`

```
def _connect_thread(self, config_file, server):
    # This blocks for 10-30 seconds
    subprocess.run(['wg-quick', 'up', config_file])
    # Signal main thread when complete
    self.connection_complete.emit(True/False)
```

PingTest Thread Pool

- **Type:** `concurrent.futures.ThreadPoolExecutor`
- **Workers:** Configurable (default 4)
- **Task:** One ping per server
- **Timeout:** 5 seconds per test
- **Results:** Aggregated and sent to main thread

Code Location: `vpn_core/speedtest_utils.py::SpeedTestManager`

```
executor = ThreadPoolExecutor(max_workers=4)
futures = [
    executor.submit(ping_server, server)
    for server in servers
]
results = [f.result(timeout=5) for f in futures]
```

Thread Safety Mechanisms

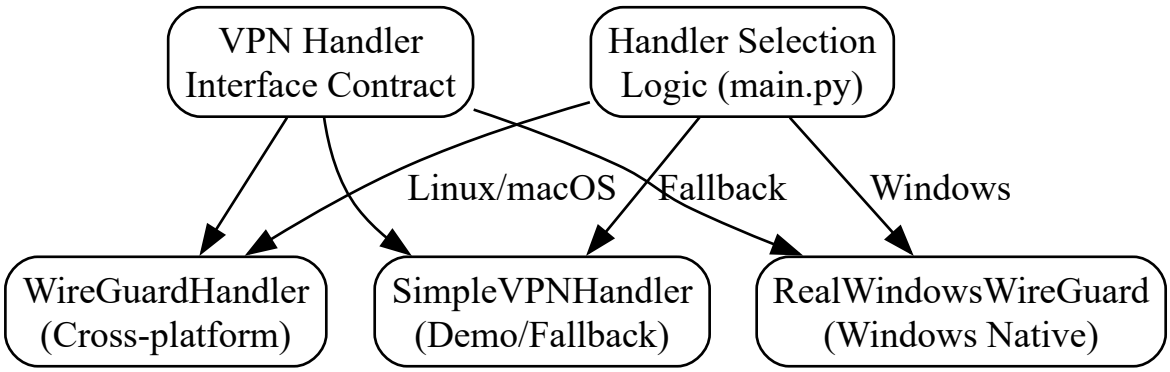
1. **Qt Signal/Slot (Automatic Queueing)**

- Signals sent from other threads → queued to main thread
- Guarantees execution order
- No deadlocks

2. **Shared Mutable State Protection**
   - `handler.is_connected`
   - `handler.current_server`
   - Protected by thread-local checks

3. **No Data Sharing Between Threads**
   - Each thread has its own data
   - Communication through signals only
   - Minimizes race conditions

Handler Polymorphism



Architecture Pattern: Strategy

The VPN handler system uses the **Strategy Pattern** to support multiple VPN implementations:

```
# Common Interface
class VPNHandlerInterface:
    def connect_to_server(self, server_id: str) -> bool: ...
    def disconnect(self) -> bool: ...
    def get_connection_status(self) -> Dict: ...
    def test_wireguard_installation(self) -> bool: ...

# Concrete Strategies
class WireGuardHandler(VPNHandlerInterface):
    """Linux/macOS implementation"""
    # Uses wg-quick command

class RealWindowsWireGuard(VPNHandlerInterface):
    """Windows-specific implementation"""
    # Delegates to WireGuard app

class SimpleVPNHandler(VPNHandlerInterface):
```

```
    """Demo/fallback implementation"""
    # Simulates VPN without requirements
```

Why This Pattern?

☑ **Extensibility:** Easy to add new handlers (MacOS-specific, WireGuard-Protocol, etc.) ☑ **Platform Flexibility:** Same codebase, different implementations ☑ **Testing:** Can swap real handler with mock ☑ **Graceful Degradation:** Falls back to demo mode if requirements missing

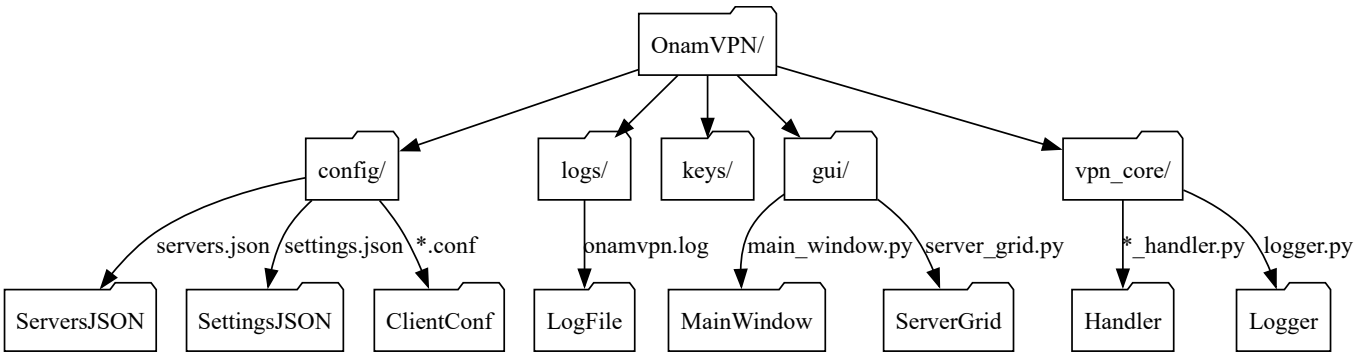
Handler Selection (main.py)

```
def select_vpn_handler():
    if platform.system() == "Windows":
        handler = RealWindowsWireGuard()
        return handler if handler.test_wireguard_installation() else
SimpleVPNHandler()
    else:  # Linux/macOS
        handler = WireGuardHandler()
        return handler if handler.test_wireguard_installation() else
SimpleVPNHandler()
```

Handler Comparison

Feature	WireGuardHandler	RealWindowsWireGuard	SimpleVPNHandler
Platforms	Linux, macOS	Windows only	All
Requires WireGuard	Yes	Yes	No
Real VPN Tunnel	Yes	Yes	No (simulated)
Automation	Full	User-guided	N/A
Speed	10-30s	(user-dependent)	2s
Code Size	420 LOC	420 LOC	320 LOC
Use Case	Production	Real Windows	Testing/Demo

File System Structure



Directory Layout

```
OnamVPN/
├── main.py                    # Entry point (260 LOC)
├── setup.py                   # Installation metadata
├── requirements.txt           # Python dependencies
├── README.md                  # User guide
├── PROJECT_EXPLANATION.md     # Architecture overview
├── run.sh                     # Linux launcher
├── run.bat                    # Windows launcher
│
├── config/                    # Configuration files
│   ├── servers.json           # Available VPN servers (static)
│   ├── settings.json          # User preferences (dynamic)
│   ├── client.conf.template   # WireGuard client template
│   ├── server.conf.template   # WireGuard server template
│   ├── {server}_client.conf   # Generated client configs
│   ├── {server}_client-{ts}.conf  # Timestamped backups
│   └── OnamVPN-Server.conf    # Server-mode config
│
├── logs/                      # Application logs (rotating)
│   ├── onamvpn.log            # Current log (active)
│   ├── onamvpn.log.1          # Previous logs (archived)
│   ├── onamvpn.log.2          # ...
│   └── onamvpn.log.5          # (5 backups total, 10MB each)
│
├── keys/                      # Temporary encryption keys
│   ├── private_key            # WireGuard private key (transient)
│   └── public_key             # WireGuard public key (transient)
│
├── gui/                       # User Interface Module
│   ├── __init__.py
│   ├── main_window.py         # MainWindow class (760 LOC)
│   ├── server_grid.py         # ServerGrid widget (320 LOC)
│   ├── settings_panel.py      # Settings dialog (780 LOC)
│   └── __pycache__/           # Compiled Python bytecode
│
└── vpn_core/                  # VPN Core Module
    ├── __init__.py
    ├── wireguard_handler.py    # Cross-platform handler (420 LOC)
    ├── real_windows_wireguard.py  # Windows handler (420 LOC)
    ├── simple_vpn_handler.py   # Demo handler (320 LOC)
    ├── speedtest_utils.py      # Ping/speed testing (270 LOC)
    ├── logger.py               # Logging setup (220 LOC)
    ├── encryption_utils.py     # AES encryption (120 LOC)
    └── __pycache__/            # Compiled Python bytecode
```

File I/O Patterns

On Startup

```
main.py
  ├─ [READ] servers.json       → Load server list
  ├─ [READ] settings.json      → Load user preferences
  ├─ [CREATE] logs/onamvpn.log → Initialize logger
  └─ [READY] for user interaction
```

On Connect

```
user.click("Connect") → handler
  ├─ [READ] servers.json                    → Get server details
  ├─ [GENERATE] wg genkey + wg pubkey        → Create keypair
  ├─ [CREATE] config/{server}_client.conf    → Save config
  ├─ [APPEND] logs/onamvpn.log               → Log connection
  ├─ [EXECUTE] wg-quick up ...                → Start tunnel
  └─ [APPEND] logs/onamvpn.log               → Log success/failure
```

On Settings Change

```
user.changes_theme() → SettingsPanel
  ├─ [READ] current settings.json
  ├─ [WRITE] settings.json with new theme
  ├─ [APPEND] logs/onamvpn.log               → Log change
  └─ [REBUILD] UI with new theme
```

On Disconnect

```
handler.disconnect()
  ├─ [EXECUTE] wg-quick down ...
  ├─ [DELETE] keys/* (if file-based)
  ├─ [APPEND] logs/onamvpn.log
  └─ [UPDATE] connection state
```

Log Rotation Policy

```
Scenario: Continuous usage (lots of connections)

File Size Limits:
  • Active log:     10 MB max
  • Backup copies:  5 total
  • Total storage:  ~50 MB max

Rotation Mechanics:
```

```
  When onamvpn.log exceeds 10 MB:
    onamvpn.log    → onamvpn.log.1
    onamvpn.log.1  → onamvpn.log.2
    ...
    onamvpn.log.4  → onamvpn.log.5
    onamvpn.log.5  → [DELETED]
    [NEW FILE] created as onamvpn.log
```

Installation & Setup

Requirements

- **Python 3.8+**
- **PySide6** (Qt for Python)
- **WireGuard** (optional, for real VPN)
- **Graphviz** (optional, for diagrams)

Installation Steps

1. Clone Repository

```
git clone https://github.com/yourusername/onamvpn.git
cd onamvpn
```

2. Install Python Dependencies

```
pip install -r requirements.txt
```

requirements.txt includes:

```
PySide6>=6.0.0
cryptography>=3.4
```

3. Install WireGuard (Optional)

Linux:

```
sudo apt-get install wireguard wireguard-tools
```

macOS:

```
brew install wireguard-tools
```

Windows:

- Download from: https://www.wireguard.com/install/
- Run installer as Administrator

4. Run Application

```
# GUI Mode (default)
python main.py

# Server Mode
python main.py --server

# Speedtest Mode
python main.py --speedtest
```

Usage Guide

Starting the Application

```
# Linux/macOS
./run.sh

# Windows
./run.bat

# Or directly
python main.py
```

GUI Walkthrough

1. Server Selection

- Displays grid of available VPN servers
- Shows real-time ping latency
- Color-coded by speed (green=fast, red=slow)
- Click a server card to select

2. Connection Control

- **Connect Button** - Initiate VPN connection
- **Disconnect Button** - Tear down tunnel

- **Status Display** - Current connection state
- **Server Info** - Selected server details

3. Settings Panel

- **Theme** - Switch between Light/Dark modes
- **Language** - English, Spanish, French, German
- **Export Settings** - Backup user preferences
- **Import Settings** - Restore from backup

4. Connection Status

- Green dot = Connected
- Red dot = Disconnected
- Yellow = Connecting/Disconnecting
- Server name displayed when connected

Typical Workflow

```
1. Launch Application
   → Shows MainWindow with server grid

2. Select a Server
   → Click on server card

3. Click "Connect"
   → Status changes to "Connecting..."
   → Wait 10-30 seconds
   → Status changes to "Connected"

4. Use VPN
   → All traffic routed through tunnel
   → Status refreshes every 2 seconds

5. Click "Disconnect"
   → Status changes to "Disconnecting..."
   → Tunnel tears down (5-10 seconds)
   → Status returns to "Disconnected"

6. Close Application
   → Settings auto-saved
   → Connection auto-closed
```

Development

Project Structure

```
# Entry point
main.py
    └─ parse arguments
    └─ select mode (GUI/Server/Speedtest)
    └─ initialize handler
    └─ launch application

# GUI Layer
gui/
    ├─ main_window.py
    │   └─ MainWindow(handler)
    │       └─ manage UI lifecycle
    │       └─ handle user events
    │       └─ monitor connection
    │
    ├─ server_grid.py
    │   └─ ServerGrid widget
    │       └─ display servers
    │       └─ ping testing
    │
    └─ settings_panel.py
        └─ SettingsDialog
            └─ theme/language
            └─ preference persistence

# VPN Core
vpn_core/
    ├─ wireguard_handler.py      (Cross-platform)
    ├─ real_windows_wireguard.py (Windows)
    ├─ simple_vpn_handler.py     (Demo)
    ├─ speedtest_utils.py        (Utilities)
    ├─ logger.py                 (Logging)
    └─ encryption_utils.py       (Crypto)

# Data Layer
config/
    ├─ servers.json   (Server list)
    └─ settings.json  (User prefs)

# Logs
logs/
    └─ onamvpn.log*   (Rotating logs)
```

Adding New Features

Adding a New Server

1. Edit `config/servers.json`
2. Add server entry with public key
3. Application auto-loads on startup

Adding a New Handler

1. Create `vpn_core/new_handler.py`
2. Implement `VPNHandlerInterface`
3. Update `main.py` selection logic
4. No UI changes needed!

Adding a New Setting

1. Add field to `SettingsPanel`
2. Update JSON schema
3. Add persistence logic
4. UI updates automatically

Testing

```
# Run specific mode
python main.py --server    # Test server mode
python main.py --speedtest # Test speedtest

# Check logs
tail -f logs/onamvpn.log

# Manual testing
# 1. Connect to a server
# 2. Verify tunnel with: wg show
# 3. Test DNS: nslookup google.com
# 4. Check IP: curl ipinfo.io
```

Troubleshooting

Connection Issues

Problem: "WireGuard not found"

- **Solution:** Install WireGuard from https://www.wireguard.com/install/
- **Fallback:** Application auto-uses demo mode

Problem: "Permission denied" on Linux

- **Solution:** `sudo usermod -a -G wireguard $USER` then logout/login
- **Or:** Run with `sudo python main.py`

GUI Issues

Problem: Theme not applying

- **Solution:** Close application, delete `settings.json`, restart

Problem: Servers not loading

- **Solution:** Check `config/servers.json` is valid JSON
- **Check logs:** `tail logs/onamvpn.log`

Performance

Problem: High CPU usage during ping

- **Solution:** Reduce thread pool size in `speedtest_utils.py`
- **Check:** `SpeedTestManager(max_workers=2)` # default is 4

Problem: Slow UI responsiveness

- **Solution:** Ensure VPN operations run in background threads
- **Check:** `ConnectionMonitor` thread is active

Contributing

1. Fork repository
2. Create feature branch: `git checkout -b feature/new-feature`
3. Make changes with tests
4. Submit pull request

License

Support

For issues, questions, or suggestions:

- **GitHub Issues:** https://github.com/yourusername/onamvpn/issues
- **Documentation:** See `system_design/` folder
- **Logs:** Check `logs/onamvpn.log` for debugging

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